

DS-GA 3001

Team Lennox

Risk-Aware Sector Rotation Bandit

Ishan Malik - im2854@nyu.edu

Maanas Lalwani - ml10092@nyu.edu

Rishit Maheshwari - rm7336@nyu.edu

A DQN agent for implied-volatility-guided sector rotation
with an algorithmic safety override

Presentation Overview

15 minutes

01	The Problem & Motivation	~1 min	02	The Data	~1 min
03	Options & Implied Volatility	~1.5 min	04	RL Framework & DQN Architecture	~2 min
05	Three Key Innovations	~2 min	06	Results & Stress Tests	~2.5 min
07	Ablation & Hyperparameter Tuning	~2 min	08	Live Demo & Q&A	~3 min

The Problem

Traditional trading bots optimize for return — and fail when it matters most.



No risk management

Standard bots maximize raw return. This can lead to drawdowns during crises, wiping out months of gains.



Backward-looking signals

Price data tells you what happened. **Implied Volatility** tells you what the market expects to happen next.



No safety boundary

Algorithms trained on normal regimes fail silently outside their training distribution.

Our solution:

RL agent using implied volatility as forward-looking state · Sortino reward shaping · Hard-coded safety override

The Data

Mixed: real market prices + VIX-based proxy for implied volatility

Real Data

Yahoo Finance

- ETF prices: XLK, XLF, XLV, SPY (2020–2024)
- Treasury yields (^IRX) as risk-free rate
- Daily log returns computed from real prices

Proxy IV

VIX-based

- XLK (Tech): $VIX \times 1.15$
- XLF (Finance): $VIX \times 1.05$
- XLV (Healthcare): $VIX \times 0.85$

1,238

Trading Days

987

Training (2020–23)

251

Test (2024)

15

Features / Day

Implied Volatility

The market's collective expectation of future movement, extracted from live options prices.

High IV

Market is fearful → options are expensive → expect large moves

Low IV

Market is calm → options are cheap → expect small moves

Why better than price?

Implied Volatility (IV) spikes BEFORE crashes — it's forward-looking, not backward

VIX Reference

< 15

Calm

20–25

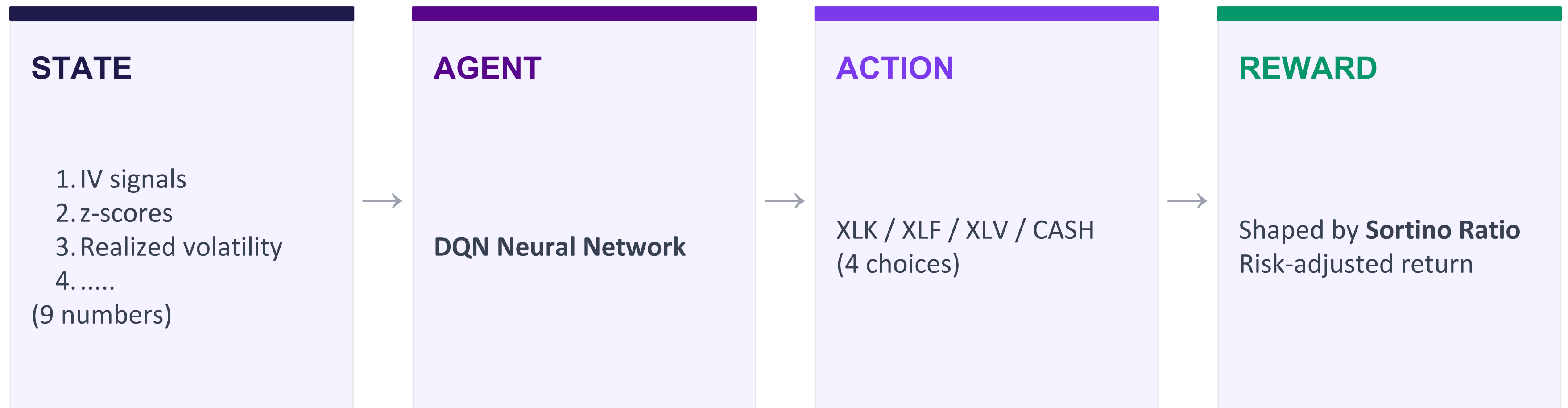
Caution

40+

Fear

Black-Scholes inversion: $C = S \cdot \Phi(d_1) - K \cdot e^{-rT} \cdot \Phi(d_2) \rightarrow$ solve for σ (IV) via Brent's algorithm

RL Framework



Bellman Update:

$$Q(s, a) \leftarrow r + \gamma \cdot \max Q(s', a')$$

2,000

Episodes
per training run

987

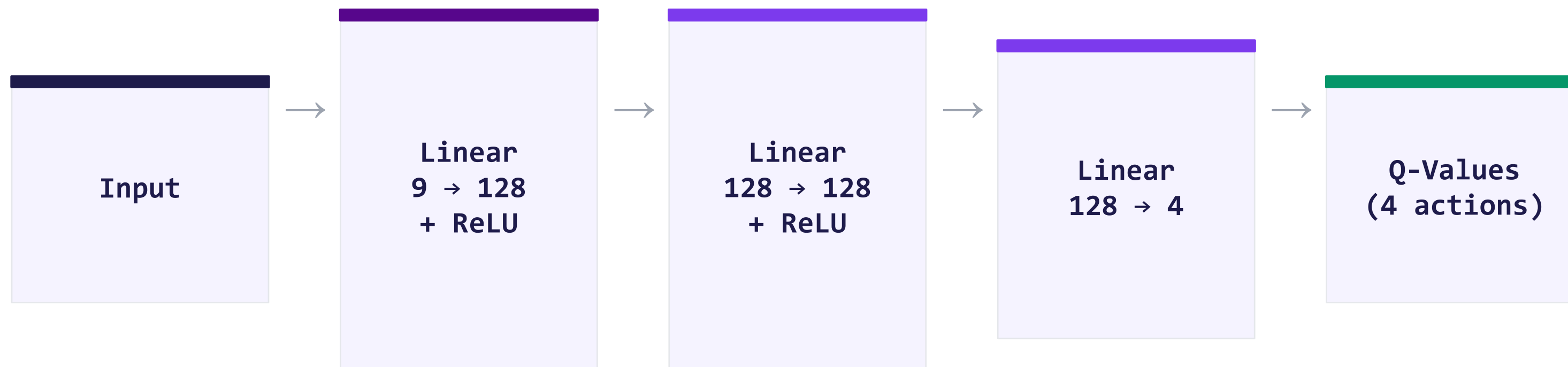
Days
per episode

$\gamma = 0.99$

Discount
factor

DQN Architecture

Discrete action space (4 sectors) → DQN is the correct choice over PPO/A2C



Experience Replay

Buffer of 10,000 transitions. Random mini-batches of 64, break temporal correlation → stable training.

Target Network

Frozen copy updated every 100 steps which provides stable Q-value targets. It even prevents oscillation.

ϵ -Greedy: 1.0 → 0.3

Starts fully random, gradually shifts toward the learned policy over 2,000 episodes.

Three Key Innovations

Each component has a measurable, independent contribution

1

IV as State Signal

Forward-looking fear measure, not backward-looking price data. During COVID (Feb 2020), IV spiked before prices crashed — the agent sees the fear signal early.

Forward-looking

2

Sortino Reward

Sharpe penalizes upside AND downside volatility. Sortino only penalizes losses. This teaches the agent to avoid drawdowns while embracing gains.

Risk-adjusted

3

Vasant Dhar Override

Hard-coded outside the gradient pathway. When ALL 3 sector IV z-scores exceed 2.5σ simultaneously → force CASH. Agent cannot learn to bypass this.

Safety layer

Training Pipeline

1 Reset environment

Return to Day 1 of training data (Jan 2020)

2 ϵ -greedy action

Select action; high ϵ early $\rightarrow$ explore;
low ϵ late $\rightarrow$ exploit

3 Environment step

Execute action $\rightarrow$ observe reward + next state

4 Store transition

Push (s, a, r, s') into replay buffer
(capacity 10K)

5 Mini-batch update

Sample 64 random transitions $\rightarrow$
compute Bellman target $\rightarrow$ update
weights

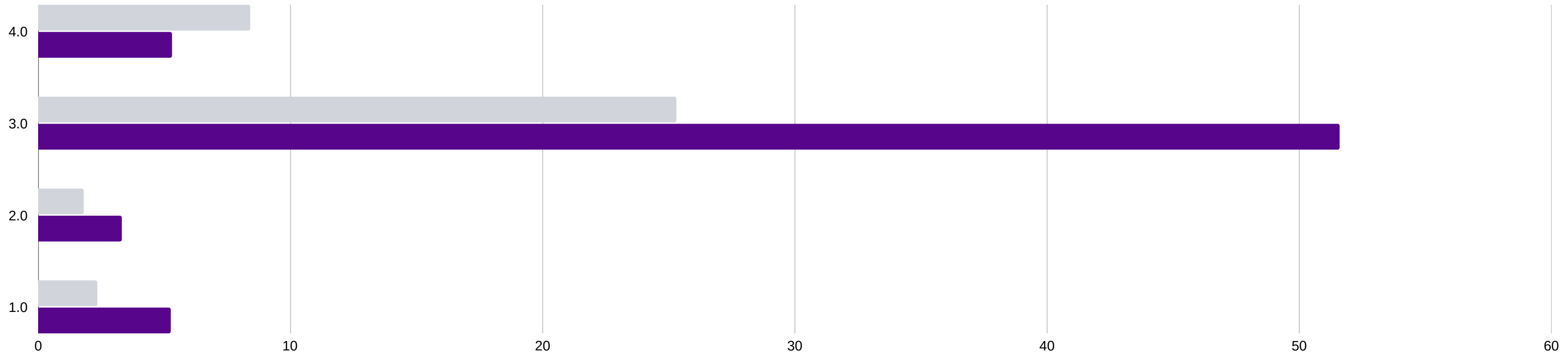
6 Validate & checkpoint

Every 50 episodes: test on 2024 data $\rightarrow$
save if best Sortino so far

MLflow $\rightarrow$ Every hyperparameter, metric, and checkpoint is logged. Experiments are fully reproducible.

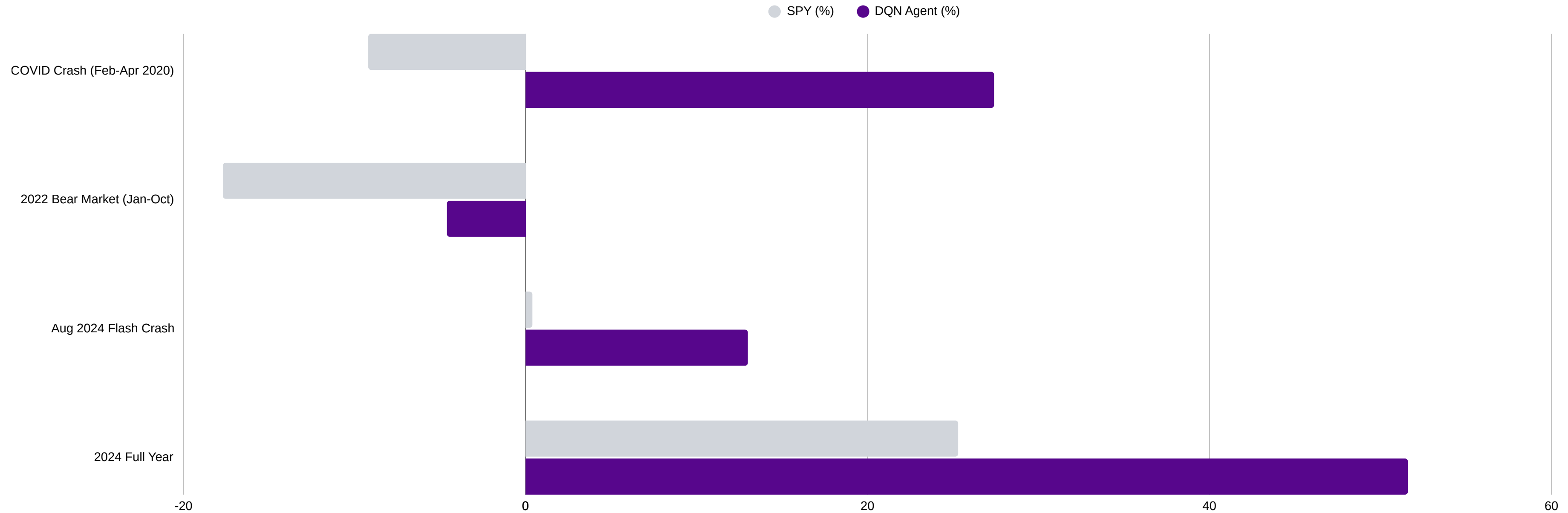
2024 Test Results

Tested on data the agent has never seen during training



Stress Tests

Agent beats SPY across all four distinct market regimes



Extreme fear → Override → CASH (COVID, Aug 2024)

Sector stress → Rotate to defensive sectors (learned behavior)

The Safety Override

"Every algorithm has an Automation Frontier — a boundary beyond which its predictions become unreliable."

— Prof. Vasant Dhar, NYU Stern (2016 HBR)

Trigger Condition

- ALL three sector IV z-scores $> 2.5\sigma$ simultaneously $\rightarrow$ FORCE CASH
- Hard-coded outside the gradient pathway. The agent cannot learn to bypass this because it operates after the Q-network makes its decision.

Historical Triggers

✓ COVID Crash

Feb–Mar 2020 · 3 triggers

✓ Aug 2024 Flash Crash

Japan carry trade · 2 triggers

✓ Total

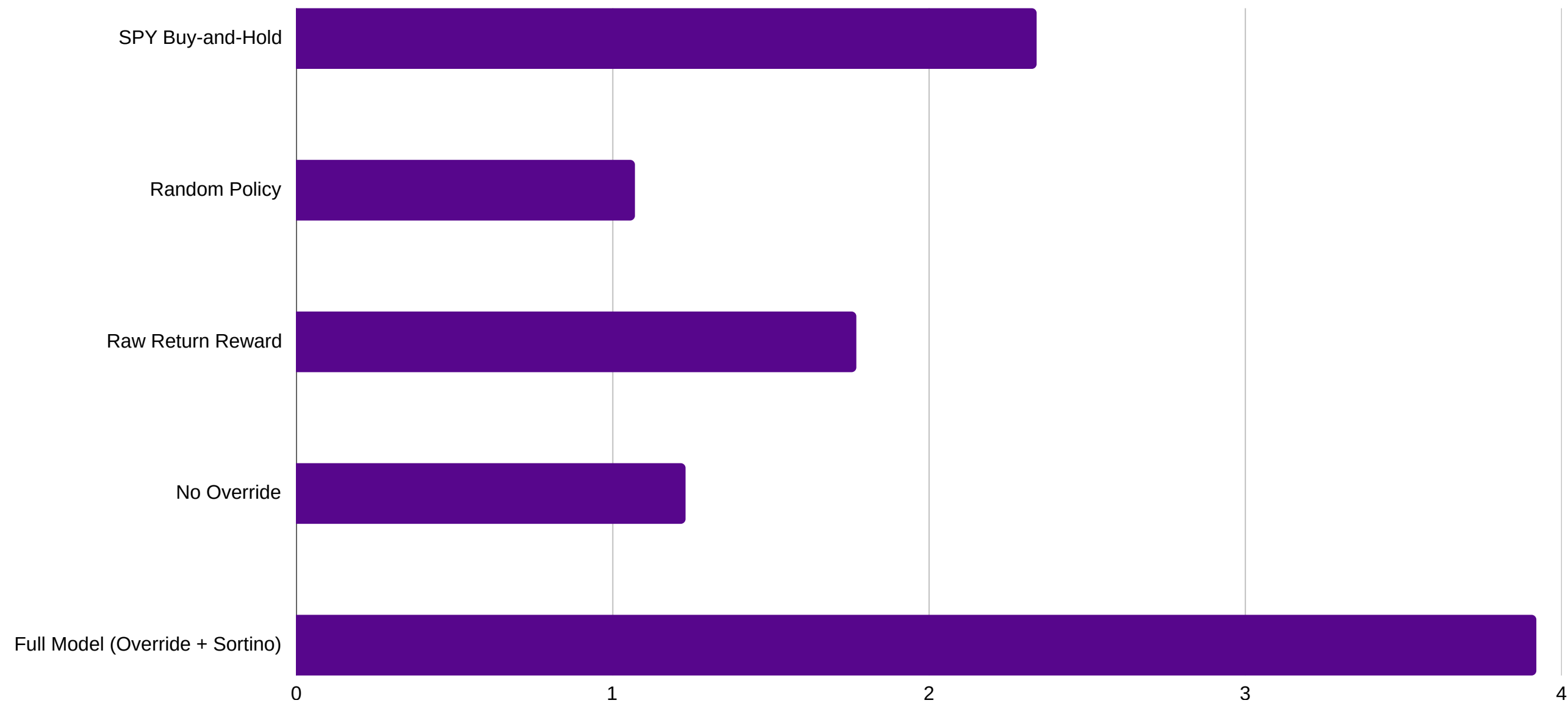
19 days · 1.5% of all trading days

Why all three sectors?

- Sector-specific fear $\rightarrow$ rotate away (learned).
- Simultaneous fear across ALL sectors $\rightarrow$ nowhere is safe $\rightarrow$ CASH.

Ablation Study

Remove one component at a time to prove each one earns its place



Remove Override

3.92 $\rightarrow$ 1.23 · MaxDD triples to -12.1%

Remove Sortino

3.92 $\rightarrow$ 1.77 · MaxDD doubles to -10.3%

Random Policy

Sortino 1.07 · Proves genuine learning occurred

Each component contributes independently — this is not just a rising-market effect.

Hyperparameter Tuning

Bayesian optimization (Optuna TPE) — learns from trials, not random grid search

Parameter	Search Space	Best Found	Why It Matters
learning_rate	[0.0001, 0.01] log-uniform	0.000103	Slow, careful weight updates → stable convergence
gamma (γ)	[0.90, 0.999] uniform	0.9807	Slightly less future-focused than default 0.99
hidden_dim	64 / 128 / 256	256	More capacity to learn complex IV–return patterns
buffer_size	[5,000 – 20,000] int	5,000	Focuses training on more recent transitions
batch_size	32 / 64 / 128	128	Larger batches, more stable gradient updates

Best trial Sortino

5.84

200-episode Optuna trial

Final model Sortino

5.25

Using original lr = 0.001

Live Demo

Tab 1 — Simulation

- Step through 2024 data day-by-day
- Real-time portfolio value + sector allocation
- Interactive charts + trading logs + **clear directives** on where to invest
- Manual Hyperparameter Selection + Option to Autonomize Agent

Tab 2 — Market

- Dataset overview (Train/Test splits, coverage, feature/action space)
- Implied Volatility (IV) and 60-day rolling IV Z-scores
- Buy-and-hold cumulative returns vs. SPY benchmark
- Realized vs. Implied Volatility (Vol Risk Premium)
- Daily return distributions across sectors

Tab 3 — Training

- Test set performance summary (Test Return, Sortino Ratio, vs. SPY)
- Final action distribution breakdown (Cash vs. Sectors)
- Live training curves (Reward per episode, Bellman loss, Epsilon decay)
- Detailed view of Agent Architecture & Hyperparameters

Tab 4 — Guide

- IV z-scores for all 3 sectors (2020–24). Project overview and team credits.
- Plain-English explanation of the Reinforcement Learning agent's mechanics
- Educational glossary defining key financial metrics (Total Return, Sortino Ratio, Max Drawdown, Alpha, etc.)

@st.cache_resource → neural network loads once per session, not on every interaction

Thank you.

Ishan Malik . im2854@nyu.edu
Maanas Lalwani . ml10092@nyu.edu
Rishit Maheshwari . rm7336@nyu.edu

NYU Center for Data Science · DS-GA 3001 · Team Lennox

5.25

Sortino Ratio

+51.6%

Total Return

4 / 4

Stress Tests

19

Safe Overrides

DQN · Implied Volatility · Sortino Reward · Vasant Dhar Override · Optuna · MLflow · Streamlit